

Qatar-1b

R-2800

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120802 · created 2026-09-04T23:52:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456141.80178 +/- 0.00016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.016 +/- 0.053
Transit depth (Rp/Rs)^2	0.026 +/- 0.17 [%]
Orbital Inclination (inc)	79.16 +/- 2.92
Airmass coefficient 1 (a1)	0.0567 +/- 0.0066
Airmass coefficient 2 (a2)	2.37 +/- 0.6
Scatter in the residuals of the lightcurve fit is	12.4 %
Best Comparison Star	#1 - [487, 55]
Optimal Aperture	1.82
Optimal Annulus	14.14
Transit Duration (day)	0.013 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.016 ± 0.053 vs 0.1463 (deviation 1.74σ)
Duration fit vs published	0.013 d vs 0.0692 d (deviation 1.9σ)
Mid-time O–C	0.4 min
Depth SNR	0.2
Residual scatter	12.4 %

Light curve

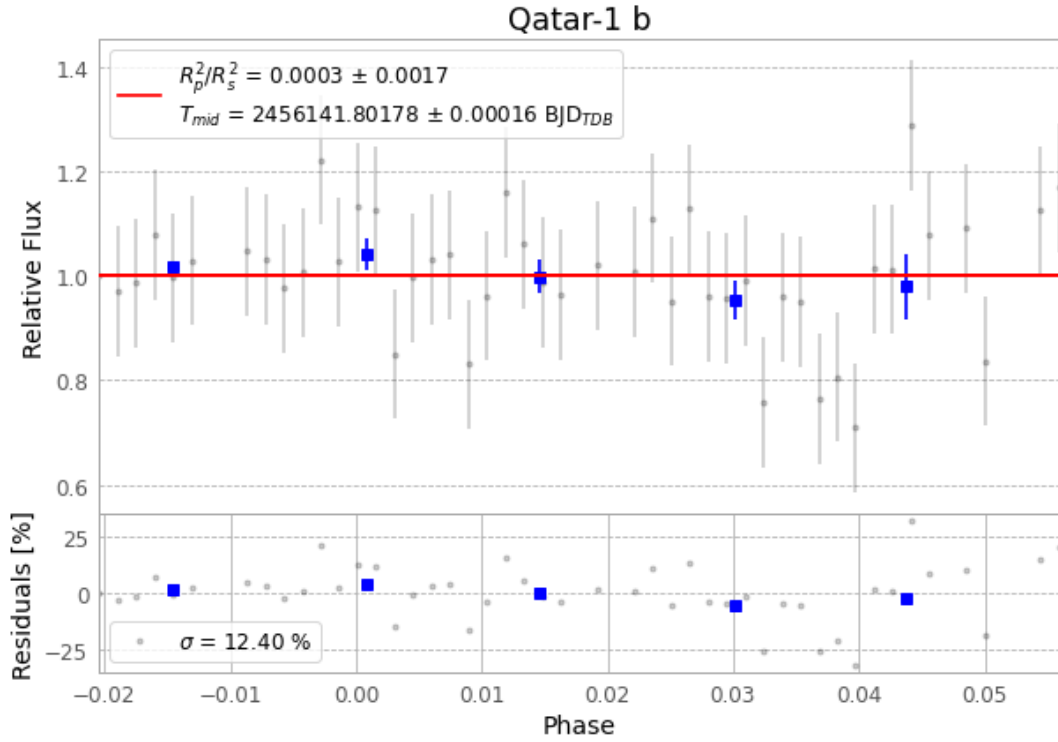


Figure 1 — FinalLightCurve_Qatar-1 b_2012-08-01.png (SHA-256 b7a4027cc8cb7b18...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [389, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0f927f5879a4ab61...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

52 FITS frames (34 MB), Qatar-1120802063012.FITS → Qatar-1120802090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120802.log (33a7e840a1f2...), FinalLightCurve_Qatar-1 b_2012-08-01.png (b7a4027cc8cb...), FinalLightCurve_Qatar-1 b_2012-08-01.csv (ecfb5dadd68d...)

Compute resources

Wall time 710.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 23:52Z.